

Complete Summary of the Handwritten computations

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Abstract

This document recounts all the calculations, derivations, and concepts covered during my hand calculations. It covers the calculation of the expectation of products of ReLU functions applied to bivariate Gaussian variables, the application of Price's theorem, the analysis of special cases, and the study of the underlying probability laws such as the Fisher and Kibble distributions, in relation to the curse of dimensionality.

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1 Calculating the Expectation $\mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)]$

Once we compute MMNN's NTK for 1 layer, we are left with the problem of computing the expectation $\mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)]$ for a pair of random variables (X_1, X_2) following a bivariate normal distribution with parameters $(\mu_1, \sigma_1, \mu_2, \sigma_2, \rho)$.

If you have 2 deterministic variables X_1 and X_2 , the output of MMNN's follow a gaussian process.

Let $\sigma_1 = \|x_1\|$, where $\|\cdot\|$ denotes the Euclidean norm. Let $\sigma_2 = \|x_2\|$, and Let $\rho = \frac{\langle x_1, x_2 \rangle}{\|x_1\| \|x_2\|}$ be the cosine similarity between x_1 and x_2 .

The covariance matrix of the gaussian process is:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

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1.1 Decomposition

We recall the expression of mu The approach is to standardize the variables ($Z_i = (X_i - \mu_i)/\sigma_i$) and decompose the integral into four terms:

$$\mathbb{E}[\dots] = \sigma_1\sigma_2 I_{22} + \mu_1\sigma_2 I_{12} + \mu_2\sigma_1 I_{21} + \mu_1\mu_2 I_{11}$$

where the integrals I_{ij} are defined over the domain $z_1 > a$ and $z_2 > b$, with $a = -\mu_1/\sigma_1$ and $b = -\mu_2/\sigma_2$.

- $I_{11} = \int_a^\infty \int_b^\infty \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{21} = \int_a^\infty \int_b^\infty z_1 \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{12} = \int_a^\infty \int_b^\infty z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$
- $I_{22} = \int_a^\infty \int_b^\infty z_1 z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$

1.2 Calculation of the Components

Calculation of I_{11} (Probability)

I_{11} is the probability $P(Z_1 > a, Z_2 > b)$. It is calculated via the cumulative distribution function (CDF) of the standard bivariate normal distribution, Φ_2 .

$$I_{11} = P(Z_1 > a, Z_2 > b) = P(Z_1 < -a, Z_2 < -b) = \Phi_2(-a, -b; \rho) = \Phi_2\left(\frac{\mu_1}{\sigma_1}, \frac{\mu_2}{\sigma_2}; \rho\right)$$

Calculation of I_{12} and I_{21} (First-Order Moments)

For I_{12} , we need to calculate:

$$I_{12} = \int_a^\infty \int_b^\infty z_2 \phi_2(z_1, z_2; \rho) dz_2 dz_1$$

Step 1: Inner integral We first focus on the inner integral with respect to z_2 . The exact formula for the conditional density is:

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(y) \phi_1\left(\frac{x - \rho y}{\sqrt{1-\rho^2}}\right)$$

Applying this formula with $x = z_1$ and $y = z_2$:

$$\phi_2(z_1, z_2; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1\left(\frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}}\right)$$

The inner integral becomes:

$$\int_b^\infty z_2 \frac{1}{\sqrt{1-\rho^2}} \phi_1(z_2) \phi_1\left(\frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}}\right) dz_2$$

Step 2: Change of variable Let $u = \frac{z_1 - \rho z_2}{\sqrt{1-\rho^2}}$. Then:

$$z_2 = \frac{z_1 - u\sqrt{1-\rho^2}}{\rho}$$

$$dz_2 = -\frac{\sqrt{1-\rho^2}}{\rho} du$$

The lower bound becomes $u_{max} = \frac{z_1 - \rho b}{\sqrt{1-\rho^2}}$. The integral transforms into:

$$\frac{1}{\sqrt{1-\rho^2}} \int_{-\infty}^{\frac{z_1 - \rho b}{\sqrt{1-\rho^2}}} \left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) \phi_1\left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) \phi_1(u) \left(-\frac{\sqrt{1-\rho^2}}{\rho}\right) du$$

Step 3: Simplification Grouping terms and using properties of ϕ_1 :

$$-\frac{1}{\rho^2} \int_{-\infty}^{\frac{z_1 - \rho b}{\sqrt{1-\rho^2}}} (z_1 - u\sqrt{1-\rho^2}) \phi_1(u) \phi_1\left(\frac{z_1 - u\sqrt{1-\rho^2}}{\rho}\right) du$$

Step 4: Using the conditional density formula The exact formula is:

$$\phi_2(x, y; \rho) = \frac{1}{\sqrt{1-\rho^2}} \phi_1(x) \phi_1\left(\frac{y - \rho x}{\sqrt{1-\rho^2}}\right)$$

Applying it correctly to our case with $x = z_1$ and $y = z_2$:

$$I_{12} = \phi_1(b) \Phi_1\left(\frac{a + \rho b}{\sqrt{1-\rho^2}}\right) + \rho \phi_1(a) \Phi_1\left(-\frac{b + \rho a}{\sqrt{1-\rho^2}}\right)$$

This final formula for I_{12} is one of the key terms in calculating the expectation of the product of ReLUs.

Calculation of I_{22} (Cross-Moment)

The calculation of I_{22} is the most complex part. The final result is:

$$I_{22} = \int_0^\rho I_{11} + \phi_2(a, b; \rho)$$

The proof of this identity was a central point of the discussion. The most rigorous method consists of showing the equality for $\rho = 0$, and then the equality of the derivatives with respect to ρ for both sides of the equation.

Final Formula

By assembling all the terms, we obtain the final algebraic formula:

$$\mathbb{E}[\dots] = (\mu_1\mu_2 + \rho\sigma_1\sigma_2)I_{11} + \mu_1\sigma_2I_{12} + \mu_2\sigma_1I_{21} + \sigma_1\sigma_2\phi_2(a, b; \rho)$$

2 Unsuccessful Attempts to Calculate I_{22}

Several approaches to calculate I_{22} proved to be dead ends.

2.1 Via the Identity on the Derivative of ϕ_2

Let's first prove the identity for the partial derivatives of the 2D normal PDF ϕ_2 . For a bivariate normal distribution with correlation ρ :

$$\frac{\partial \phi_2}{\partial z_1} = -\frac{z_1 - \rho z_2}{1 - \rho^2} \phi_2(z_1, z_2; \rho)$$

$$\frac{\partial \phi_2}{\partial z_2} = -\frac{z_2 - \rho z_1}{1 - \rho^2} \phi_2(z_1, z_2; \rho)$$

Using these identities, we can write:

$$z_1 \phi_2 = -\rho z_2 \phi_2 - (1 - \rho^2) \frac{\partial \phi_2}{\partial z_1}$$

Integrating from a to ∞ with respect to z_1 :

$$\int_a^\infty z_1 \phi_2 dz_1 = -\rho z_2 \int_a^\infty \phi_2 dz_1 + (1 - \rho^2) \phi_2(a, z_2; \rho)$$

Attempting to use this for I_{22} , we integrate $z_1 z_2 \phi_2$ over the region $[a, \infty) \times [b, \infty)$:

$$I_{22} = \int_b^\infty z_2 \left(\int_a^\infty z_1 \phi_2 dz_1 \right) dz_2$$

Substituting the identity:

$$I_{22} = -\rho \int_b^\infty z_2^2 \left(\int_a^\infty \phi_2 dz_1 \right) dz_2 + (1 - \rho^2) \int_b^\infty z_2 \phi_2(a, z_2; \rho) dz_2$$

This approach becomes intractable because: 1. The first term involves a higher-order moment z_2^2 2. The second term requires evaluating another complex integral 3. Each attempt to simplify leads to terms of equal or greater complexity

Therefore, while the partial derivative identities are mathematically elegant, they do not provide a practical path to computing I_{22} when ρ is non 0.

3 Special Case: Zero Means ($\mu_1 = \mu_2 = 0$)

In this case, the "from scratch" derivation leads to an elegant formula.

$$\mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)] = \sigma_1 \sigma_2 \int_0^\infty \int_0^\infty z_1 z_2 \phi_2(z_1, z_2; \rho) dz_1 dz_2$$

The problem reduces to calculating $I_{22}(0, 0; \rho) = \mathbb{E}[\text{ReLU}(Z_1) \text{ReLU}(Z_2)]$. The cleanest derivation uses Price's theorem:

1. $\frac{\partial \mathbb{E}[g]}{\partial \rho} = \mathbb{E} \left[\frac{\partial^2 g}{\partial z_1 \partial z_2} \right] = \mathbb{E}[\mathbb{I}(z_1 > 0, z_2 > 0)] = P(Z_1 > 0, Z_2 > 0) = \frac{1}{2\pi} \arccos(-\rho)$.
2. The integration constant is $\mathbb{E}[g]_{\rho=0} = \mathbb{E}[\text{ReLU}(Z_1)]\mathbb{E}[\text{ReLU}(Z_2)] = \left(\frac{1}{\sqrt{2\pi}}\right)^2 = \frac{1}{2\pi}$.
3. By integrating $\frac{\partial \mathbb{E}[g]}{\partial \rho}$ from 0 to ρ :

$$\begin{aligned} \mathbb{E}[g](\rho) - \mathbb{E}[g](0) &= \int_0^\rho \frac{1}{2\pi} \arccos(-t) dt \\ &= \frac{1}{2\pi} \left[t \arccos(-t) + \sqrt{1-t^2} \right]_0^\rho \\ &= \frac{1}{2\pi} (\rho \arccos(-\rho) + \sqrt{1-\rho^2} - 1) \end{aligned}$$

4. By adding the constant $\mathbb{E}[g](0) = 1/(2\pi)$, we get:

$$\mathbb{E}[\text{ReLU}(Z_1) \text{ReLU}(Z_2)] = \frac{1}{2\pi} (\sqrt{1-\rho^2} + \rho \arccos(-\rho))$$

The final formula for variances σ_1^2, σ_2^2 is:

$$\mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)] = \frac{\sigma_1 \sigma_2}{2\pi} (\sqrt{1-\rho^2} + \rho \arccos(-\rho))$$

4 Price's Theorem and Derivations

Price's theorem was a central tool in our analysis.

Statement of price's theorem

Suppose that X_1 and X_2 forms a gaussian vector with joint distribution $f_{X_1, X_2}(x_1, x_2)$. Then, for any function $g(x_1, x_2)$, we have:

$$\mathbb{E}[g(X_1, X_2)] = \int_0^\rho \mathbb{E} \left[\frac{\partial^2 g}{\partial x_1 \partial x_2}(X_1, X_2) \right] d\rho + \text{Constant}$$

where the constant is the value of the expectation when $\rho = 0$.

4.1 Explicit Formulas for Σ and $\dot{\Sigma}$

For layer ℓ , the general formulas are:

$$\begin{aligned} \Sigma^{(\ell)}(x, x') &= \mathbb{E}_{w, b} \left[\sigma(w^\top h^{(\ell-1)}(x) + \beta b) \sigma(w^\top h^{(\ell-1)}(x') + \beta b) \right] \\ \dot{\Sigma}^{(\ell)}(x, x') &= \mathbb{E}_{w, b} \left[\dot{\sigma}(w^\top h^{(\ell-1)}(x) + \beta b) \dot{\sigma}(w^\top h^{(\ell-1)}(x') + \beta b) w^\top w \right] \end{aligned}$$

When $\beta = 0$, these simplify to:

$$\begin{aligned} \Sigma^{(\ell)}(x, x') &= \mathbb{E}_w \left[\sigma(w^\top h^{(\ell-1)}(x)) \sigma(w^\top h^{(\ell-1)}(x')) \right] \\ \dot{\Sigma}^{(\ell)}(x, x') &= \mathbb{E}_w \left[\dot{\sigma}(w^\top h^{(\ell-1)}(x)) \dot{\sigma}(w^\top h^{(\ell-1)}(x')) w^\top w \right] \end{aligned}$$

Note that for $\ell > 1$, both kernels become random fields due to the randomness in $h^{(\ell-1)}$.

4.2 Differential and Integral Form

The theorem relates the derivative of the expectation of a function g with respect to the covariance σ_{12} to the expectation of the second derivative of g . For $g = \text{ReLU}(X_1) \text{ReLU}(X_2)$, we have $\frac{\partial^2 g}{\partial x_1 \partial x_2} = \mathbb{I}(X_1 > 0, X_2 > 0)$.

$$\frac{\partial \mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)]}{\partial \sigma_{12}} = \mathbb{E}[\mathbb{I}(X_1 > 0, X_2 > 0)] = I_{11}$$

This leads to the integral formulation:

$$\mathbb{E}[\text{ReLU}(X_1) \text{ReLU}(X_2)] = \int I_{11}(\sigma_{12}) d\sigma_{12} + \text{Constant}$$

4.3 Calculating the Constant (Independent Case $\rho = 0$)

The constant is the value of the expectation when $\rho = 0$, i.e., when X_1 and X_2 are independent.

$$C = \mathbb{E}[\text{ReLU}(X_1)] \cdot \mathbb{E}[\text{ReLU}(X_2)]$$

The expectation of a single ReLU is:

$$\mathbb{E}[\text{ReLU}(X)] = \sigma \phi_1 \left(\frac{\mu}{\sigma} \right) + \mu \Phi_1 \left(\frac{\mu}{\sigma} \right)$$

The constant is therefore the product of two such terms:

$$C = \left[\sigma_1 \phi_1 \left(\frac{\mu_1}{\sigma_1} \right) + \mu_1 \Phi_1 \left(\frac{\mu_1}{\sigma_1} \right) \right] \times \left[\sigma_2 \phi_1 \left(\frac{\mu_2}{\sigma_2} \right) + \mu_2 \Phi_1 \left(\frac{\mu_2}{\sigma_2} \right) \right]$$

This is the implementation of the constant in code, this is fairly long for non zeros beta, that's why we focus on the hypothesis with beta = 0. This in order to make experiments and proofs that MMNNs are superior to FCNNs for 2 hidden layer networks.

5 MMNN NTK for 2 hidden layers

5.1 recursive

For a 2 hidden layer MMNN, we can write the recursive formulation of the NTK kernel. Let's recall the general recursive formula for layer ℓ :

suppose that you have a low rank hidden layer, with a width r
so the architecture is :

$$\begin{array}{ccccccc} (x, y) & \xrightarrow{\sigma W_1} & h_1 & \xrightarrow{A_1 \cdot + c_1} & r & \xrightarrow{\sigma W_2} & h_2 \xrightarrow{A_2 \cdot + c_2} (x_2, y_2) \\ & & N & & (x_1, y_1) \text{ in } \mathbb{R}^r & & N \end{array}$$

And here N grow to infinity, and r is fixed.

Then the NTK random kernel is:

$$\Theta^{(\ell)}(x, x') = \Theta^{(\ell-1)}(x, x') \cdot \left(\sigma_A^2 \dot{\Sigma}^{(\ell)}(x, x') \right) + \Sigma^{(\ell)}(x, x') + 1 \quad (1)$$

For $\ell = 2$, starting from $\Theta^{(0)} = 0$, we have:

$$\begin{aligned} \Theta^{(1)} &= \mathbb{E}_{w,b} \left[\sigma(w^\top x + \beta b) \sigma(w^\top x' + \beta b) \right] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_{w,b} \left[\dot{\sigma}(w^\top h^{(1)}(x) + \beta b) \dot{\sigma}(w^\top h^{(1)}(x') + \beta b) w^\top w \right] \right) \\ &\quad + \mathbb{E}_{w,b} \left[\sigma(w^\top h^{(1)}(x) + \beta b) \sigma(w^\top h^{(1)}(x') + \beta b) \right] + 1 \end{aligned}$$

Here, $\Theta^{(1)}$ is a deterministic NNGP kernel (Gaussian Process) since it depends only on the input x , while $\Theta^{(2)}$ is a random field (Deep Gaussian Process) that depends on the realization of the first layer's output $h^{(1)}$. When $\beta = 0$, these expectations simplify by removing the bias terms b . This non-Gaussian nature of the kernels for $\ell > 1$ is a key distinguishing feature of MMNNs compared to standard neural networks.

When $\beta = 0$, the equations simplify to:

Main focus: NTK kernels with β

$$\begin{aligned} \Theta^{(1)} &= \mathbb{E}_w \left[\sigma(w^\top x) \sigma(w^\top x') \right] + 1 \\ \Theta^{(2)} &= \Theta^{(1)} \cdot \left(\sigma_A^2 \mathbb{E}_w \left[\dot{\sigma}(w^\top h^{(1)}(x)) \dot{\sigma}(w^\top h^{(1)}(x')) w^\top w \right] \right) \\ &\quad + \mathbb{E}_w \left[\sigma(w^\top h^{(1)}(x)) \sigma(w^\top h^{(1)}(x')) \right] + 1 \end{aligned}$$

Note that for stable signal propagation at the Edge of Chaos (EOC), we require for 1 rank hidden layer:

$$\sigma_A^2 \cdot \text{Tr}(\Sigma_w) = 2$$

Then for a rank r

$$\sigma_A^2 \cdot \text{Tr}(\Sigma_w) = \frac{2}{r}$$

And we assume at least isotropic gaussian for the weights without loss of generality and to ensure fast computations.

This condition ensures the signal neither vanishes nor explodes as it propagates through the network layers. This also means we need to normalize by $\sqrt{r}$ the output of the first MMNN layer. We call x_1 and y_1 the outputs of the first layer unnormalized.

We recall also that $\Sigma^{(1)}$ is the arccosine kernel, which has a closed form in function of the correlation ρ between x and x' :

$$\Sigma^{(1)}(\rho) = \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + \rho(1 - \arccos(\rho)) \right)$$

We recall also that $\dot{\Theta}^{(1)}(\rho) = \mathbb{E}_w [w^\top w \dot{\sigma}(w^\top x) \dot{\sigma}(w^\top x')]$. We can compute this exactly when w is isotropic gaussian with variance 1:

In fact a simple computation takes the expectation of the product of the two derivatives of the ReLU, by independence of $\|w\|^2$ and $\mathbb{K}_{w^\top x > 0} \cdot \mathbb{K}_{w^\top x' > 0}$:

$$\begin{aligned} \dot{\Theta}^{(1)}(\rho) &= \mathbb{E}_w \left[w^\top w \dot{\sigma}(w^\top x) \dot{\sigma}(w^\top x') \right] \\ &= \mathbb{E}_w \left[w^\top w \cdot \mathbb{K}_{w^\top x > 0} \cdot \mathbb{K}_{w^\top x' > 0} \right] \\ &= \mathbb{E}[\|w\|^2] \cdot \mathbb{P}(w^\top x > 0, w^\top x' > 0) \\ &= \text{Tr}(\Sigma_w) \left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \right) \end{aligned}$$

It is just a reformulation of the fact that we integrate over a cone of angle $\arccos(\rho)$ a radial function.

We then write $\Theta^{(2)}$ as a function of ρ and ρ_1 :

$$\Theta^{(2)}(\rho, \rho_1) = \Theta^{(1)}(\rho) \cdot \sigma_A^2 \text{Tr}(\Sigma_w) \left(\frac{1}{2} - \frac{\arccos(\rho)}{2\pi} \right) + \frac{1}{r} \cdot \Sigma^{(1)}(\rho_1) \|x_1\| \|y_1\| + 1$$

with $\Sigma^{(1)}(\rho) = \frac{1}{\pi} \left(\sqrt{1 - \rho^2} + \rho(1 - \arccos(\rho)) \right)$

where ρ_1 is the correlation between the outputs of the first layer, that is a random variable and ρ is the correlation between the inputs.

Note that this multiplicative value is between 0 and 1, that is we can compute a lot of stuff after concatenating layers and getting some exponential scaling laws. We emphasize the fact we multiply by $\|x_1\| \|y_1\|$ which is a random variable, but as we will see, it is independent of the correlation ρ_1 .

5.2 Expression for random variables ρ_1 , $\|x_1\|^2$, $\|y_1\|^2$

The output of the 1st layer is a gaussian process of dimension r , where each coordinate is a gaussian process with covariance matrix:

$$\Sigma_i^{(1)} = \mathbb{E}_w \left[\sigma(w^\top x) \sigma(w^\top x') \right]$$

and the coordinates are ρ -correlated. If we stack horizontally the outputs x_1 and y_1 in a vector of dimension $2r$, the full $2r \times 2r$ covariance matrix is therefore:

$$\Sigma^{(1)} = \begin{pmatrix} I & \rho I \\ \rho I & I \end{pmatrix}$$

It's a kronecker product of r matrices I and 2d ρ correlated gaussian variables.

it happens, by a magic, that we can compute exactly the joint distribution of the random variable $(\rho_1, \|x_1\|^2, \|y_1\|^2)$:

$$\rho_1 = \frac{\sum_{i=1}^r x_i y_i}{\sqrt{\sum_{i=1}^r x_i^2} \sqrt{\sum_{i=1}^r y_i^2}}$$

The squared norms x_1^2 and y_1^2 follow a bivariate Kibble distribution and their cosine distance follows an independent Fisher like distribution. The Kibble distribution is defined for $u, v \geq 0$ and $\rho \in [-1, 1]$, while the Fisher distribution is defined for $\theta \in [-1, 1]$ and $\rho \in [-1, 1]$ with $r > 2$.

$$\begin{aligned}
(\|x_1\|^2, \|y_1\|^2) &\sim \text{Kibble}(r, \rho) \\
&= \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1} \rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1} \left(\frac{\rho\sqrt{uv}}{1-\rho^2} \right) \\
\rho_1 &\sim \text{Fisher}(\rho, r) \\
&= \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-\theta^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho\theta)^{r-\frac{3}{2}}} \times {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; \frac{1+\rho\theta}{2} \right)
\end{aligned}$$

where $u = \|x_1\|^2$, $v = \|y_1\|^2$, $\theta = \cos(x_1, y_1)$, and I_α is the modified Bessel function of the first kind.

and the correlation of x, y :

where x_{1i} and y_{1i} are ρ -correlated gaussian variables.

We ensure that the reader has understood that the correlation of x, y is independent of the squared norms of x, y !

5.2.1 Final result

We can then state the final result: For 2 entries x, y in $\mathbb{R}^d$ and a low rank r hidden layer, the NTK for a 2 hidden layer MMNN is:

$$\begin{aligned}
\Theta^{(2)}(x, y) &= \Theta^{(1)}(\rho) \cdot \left(1 - \frac{\arccos(\rho_1)}{\pi} \right) \frac{1}{r} + \frac{1}{r} \Sigma^{(1)}(\rho_1) \|x_1\| \|y_1\| + 1 \\
\text{with } \Sigma^{(1)}(\rho) &= \frac{1}{\pi} \left(\sqrt{1-\rho^2} + \rho(1-\arccos(\rho)) \right)
\end{aligned}$$

where the random variables follow:

- $\rho_1 \sim \text{Fisher}(\rho, r)$ is independent of the norms
- $(\|x_1\|^2, \|y_1\|^2) \sim \text{Kibble}(r, \rho)$ are chi-squared variables with r degrees of freedom from correlated gaussian variables.

6 Fisher, Kibble distributions and the fake "Curse of Dimensionality"

6.1 Distribution of the Correlation Coefficient r

Fisher derived the exact law of r in 1915, representing the cosine between Gaussian vectors of dimension r with correlation ρ :

$$p(r|\rho, r) = \frac{(r-2)\Gamma(r-1)(1-\rho^2)^{\frac{r-1}{2}}(1-r^2)^{\frac{r-4}{2}}}{\sqrt{2\pi}\Gamma(r-\frac{1}{2})(1-\rho r)^{r-\frac{3}{2}}} \times {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; r - \frac{1}{2}; \frac{1+\rho r}{2}\right)$$

Where Γ is the gamma function and ${}_2F_1$ is the hypergeometric function. For large r , Fisher's z-transform gives $z = \operatorname{arctanh}(r) \sim \mathcal{N}\left(\operatorname{arctanh}(\rho) + \frac{\rho}{2(r-1)}, \frac{1}{r-3}\right)$. The variance of r decreases as $O(1/r)$. When $\rho \sim O(1/\sqrt{r})$, geometric noise masks correlation.

6.2 The Kibble Distribution

The joint law of correlated chi-squared variables $U = \|X\|^2$ and $V = \|Y\|^2$ follows Kibble's 1941 distribution:

$$f(u, v) = \frac{(uv)^{\frac{r/2-1}{2}} e^{-\frac{u+v}{2(1-\rho^2)}}}{\Gamma(r/2)(2(1-\rho^2))^{r/2+1} \rho^{\frac{r/2-1}{2}}} \cdot I_{r/2-1}\left(\frac{\rho\sqrt{uv}}{1-\rho^2}\right)$$

Where Γ is the gamma function and I_α is the modified Bessel function of the first kind. Mean $\mathbb{E}[U] = r$ and variance $\operatorname{Var}(U) = 2r$ grow linearly. U/r converges to $\mathcal{N}(1, 2/r)$ with polynomial concentration.

6.3 Exact moments and correlation for Kibble variables

Let $(X_i, Y_i)_{i=1}^r$ be i.i.d. Gaussian pairs with zero mean, unit variances and correlation ρ , and define

$$U = \sum_{i=1}^r X_i^2, \quad V = \sum_{i=1}^r Y_i^2.$$

Then (U, V) follows Kibble's bivariate chi-square law with r degrees of freedom and correlation ρ . By Isserlis' (Wick's) theorem (or by expanding $\mathbb{E}[(X_i - \rho Y_i)^n (Y_i)^m]$ recursively):

$$\mathbb{E}[X_i^2] = 1, \quad \operatorname{Var}(X_i^2) = 2, \quad \mathbb{E}[X_i^2 Y_i^2] = 1 + 2\rho^2, \quad (X_i^2, Y_i^2) = 2\rho^2.$$

Using independence across i :

$$\mathbb{E}[U] = r, \quad \operatorname{Var}(U) = 2r, \quad \mathbb{E}[V] = r, \quad \operatorname{Var}(V) = 2r,$$

$$(U, V) = \sum_{i=1}^r (X_i^2, Y_i^2) = 2r \rho^2,$$

$$(U, V) = \frac{(U, V)}{\sqrt{\operatorname{Var}(U) \operatorname{Var}(V)}} = \frac{2r \rho^2}{\sqrt{(2r)(2r)}} = \rho^2,$$

$$\mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r \rho^2.$$

6.4 Product of Square Roots of Kibble Variables

Let $S = \sqrt{U}\sqrt{V} = \|X\|\|Y\|$. Due to correlation between U and V , we cannot directly multiply expectations. Instead, using Cauchy-Schwarz inequality:

$$\mathbb{E}[S^2] = \mathbb{E}[UV] = \mathbb{E}[U]\mathbb{E}[V] + (U, V) = r^2 + 2r\rho^2$$

Therefore:

$$\mathbb{E}[S] \leq \sqrt{\mathbb{E}[S^2]} = r\sqrt{1 + \frac{2\rho^2}{r}}$$

The variance can be estimated similarly by expanding $(S - \mathbb{E}[S])^2$ and using correlation terms:

$$\text{Var}(S) \approx \frac{r}{2}(1 + \rho^2) + O(1)$$

This suggests S/r approaches a distribution with mean $O(\sqrt{1 + \frac{2\rho^2}{r}})$ and variance of order $O(1/r)$.

6.5 Independence Properties and Asymptotic Behavior

The orientation variable r is independent of magnitudes (U, V) . For the normalized product S/r , we have:

$$\frac{S}{r} = \frac{\|X\|\|Y\|}{r} \xrightarrow{r \rightarrow \infty} 1$$

This convergence occurs with fluctuations of order $O(1/\sqrt{r})$, showing that in high dimensions, the normalized product concentrates around 1 regardless of the correlation ρ .